Cloud Computing & DevOps – AWS Handbook

SESSION 4

AWS Lambda & Serverless Computing

Building Applications Without Managing Servers

Duration	2 Hours
Prerequisites	AWS Account, Basic Programming Concepts
Learning Outcome	Create Lambda Functions, Execute Serverless Code, Understand Event-Driven Computing, Triggers, and Serverless Architecture.

Why Are We Learning This?

Traditional applications require servers to be provisioned, patched, monitored and maintained. AWS Lambda introduces the concept of serverless computing where developers focus only on application logic while AWS manages the infrastructure.

Architecture Diagram

User/Event
↓
Trigger
↓
AWS Lambda Function
↓
Code Execution
↓
Response

Key Concepts

Serverless Computing	Run code without managing servers.
Lambda Function	Unit of execution containing application logic.
Event	Action that invokes a Lambda function.
Trigger	AWS service or event source that starts execution.
Runtime	Programming environment used by Lambda.
CloudWatch Logs	Service used to view execution logs.

Practical Activities

Activity 1 – Create a Lambda Function

Expected Result: Function created successfully.

Activity 2 – Write Simple Function Logic

Expected Result: Code saved without errors.

Activity 3 – Configure a Test Event

Expected Result: Event ready for execution.

Activity 4 – Execute the Function

Expected Result: Successful execution response.

Activity 5 – Review Output and Logs

Expected Result: Logs visible in CloudWatch.

Expected Outputs

Lambda Function created, test event configured, successful execution response generated, and execution logs available for review.

What You Should See

Paste Screenshot Here

Lambda Function Dashboard

Paste Screenshot Here

Function Code Editor

Paste Screenshot Here

Test Event Configuration

Paste Screenshot Here

Successful Function Execution

Paste Screenshot Here

CloudWatch Logs Output

Journal Writing Template

Aim

To create and execute a serverless function using AWS Lambda.

Theory

AWS Lambda enables developers to execute code without managing servers and supports event-driven applications.

Procedure

Write 5–7 concise steps performed during the practical session.

Observations

Function was created successfully, event execution completed successfully, and logs were generated.

Conclusion

AWS Lambda was used successfully to execute application logic in a serverless environment.

Viva Preparation

1. What is AWS Lambda?
2. What is Serverless Computing?
3. What is an Event?
4. What is a Trigger?
5. What are the advantages of Serverless Architecture?
6. What is CloudWatch?
7. Difference between EC2 and Lambda?

Troubleshooting Box

Problem	Possible Cause
Function not executing	Configuration or trigger issue
Execution failed	Programming logic error
Permission denied	IAM role issue
No logs visible	CloudWatch configuration problem

Industry Relevance

AWS Lambda is heavily used for automation, backend APIs, file processing, notifications, event-driven workflows, and cloud-native application development. It helps organizations reduce operational overhead and pay only for actual execution time.

Session Summary

Students learned how to create Lambda functions, configure events, execute serverless code, analyze output, and understand the fundamentals of serverless computing.